

FogTorch II:

1) Estimating Resource/Consumption & Cost Estimation.

Requirement

Component

Given:

Fog Node A:

RAM = 10GB

HDD = 60GB

3 CPU's

RAM = 2GB

HDD = 55GB.

Fog Node B:

RAM = 20GB

HDD = 80GB.

~~Fog Node C:~~

~~RAM = 30GB~~

~~HDD = 60GB~~

2) Resource Consumption.

$$= \frac{1}{2} \left(\frac{\sum_{\gamma \in A} \text{RAM}(\gamma)}{\sum_{f \in F} \text{RAM}(f)} + \frac{\sum_{\gamma \in A} \text{HDD}(\gamma)}{\sum_{f \in F} \text{HDD}(f)} \right)$$

$$\text{RAM Consumption} = \frac{2}{10+20} = \frac{2}{30} = 0.066 \text{ (6.6\%)}$$

$$\text{HDD Consumption} = \frac{55}{60+80} = \frac{55}{140} = 0.3571 \text{ (35.71\%)}$$

$$\text{Total Resource Consumption} = \frac{1}{2} \times (0.066 + 0.3571)$$

$$= \frac{0.4231}{2}$$

$$= 0.21155$$

$$= 21.15\%$$

2) cloud 1 $\Rightarrow$ Monthly cost $\Rightarrow$ \$80000
 On demand $\Rightarrow$ \$1000/CPU
 \$2000/GB : RAM
 \$4000/GB : HDD

$$\text{Cost estimator} = \begin{cases} P(H, n) = \\ \sum_{PER} [H \cdot \text{exc}(e, n)] \end{cases} \quad \begin{cases} c(H, n) \end{cases}$$

Estimate the cost for above Requirement.

a) Default VM: \$80000.

b) On-Demand VM = $(3 \times 1000) + (2 \times 2000) + (55 \times 4000)$.

$$= 3000 + 4000 + 220000$$

$$= \$227000$$

$\therefore$ The cost effective solution is to opt Default VM with monthly cost \$80000